

# SpringBoot

## What is Spring Boot ?

- Spring Boot is a Java framework built on top of the Spring Framework.
- It simplifies application development by removing complex configurations.
- It provides an embedded server like Tomcat, so apps run independently.
- It helps developers build web applications and REST APIs faster with minimal code.

## Why learn Spring Boot?

1. **Fast Development:** Quickly build production ready apps with minimal configuration.
2. **Auto-Configuration:** Reduces boilerplate code, you can focus on business logic
3. **Microservices Friendly:** Perfect for building and managing distributed systems.
4. **Seamless Integrations:** Works easily with databases, security, messaging and cloud platforms.
5. **Wide Adoption:** Backed by Spring ecosystem and used in real world enterprise applications.

## Key Features of Spring Boot :-

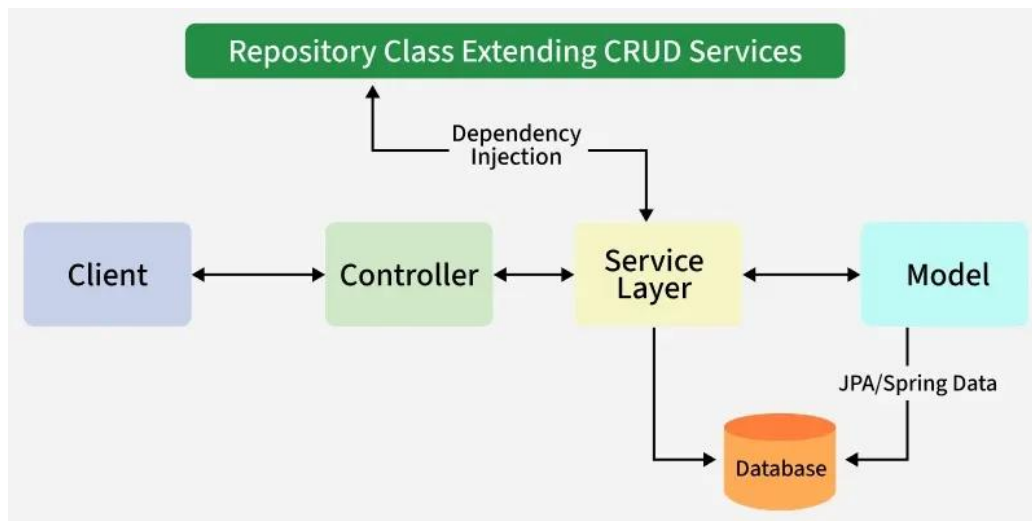
1. **Auto-Configuration** – Automatically sets up required components based on your project.
2. **Starter Dependencies** – Provides ready-made dependency bundles for faster setup.
3. **Embedded Servers** – Runs directly with built-in Tomcat/Jetty without external deployment.
4. **Production-Ready Features** – Includes Actuator for monitoring and health checks.
5. **Less Configuration** – Reduces XML and boilerplate code using defaults.
6. **DevTools Support** – Enables auto-restart and live reload for faster development.
7. **Microservices Friendly** – Makes it easy to develop and deploy microservices.

## Applications of Spring Boot :-

1. **Enterprise Applications:** Helps build large systems like hospital, banking, and ERP applications.
2. **Cloud-Native Apps:** Works smoothly with Docker, Kubernetes, and cloud platforms for easy deployment.
3. **Real-Time Applications:** Supports chat apps, streaming platforms, IoT, and event-driven systems.
4. **Batch Processing:** Used for ETL tasks, report generation, and handling large data batches.

## Architecture of Spring Boot :-

To understand the architecture of Spring Boot, let's examine its different layers and components.



### 1. Client Layer -

- This represents the external system or user that interacts with the application by sending HTTPS requests.

### 2. Controller Layer (Presentation Layer) -

- Handles incoming HTTP requests from the client.
- Processes the request and sends a response.
- Delegates business logic processing to the Service Layer.

### 3. Service Layer (Business Logic Layer) -

- Contains business logic and service classes.
- Communicates with the Repository Layer to fetch or update data.
- Uses Dependency Injection to get required repository services.

### 4. Repository Layer (Data Access Layer) -

- Handles CRUD (Create, Read, Update, Delete) operations on the database.
- Extends Spring Data JPA or other persistence mechanisms.

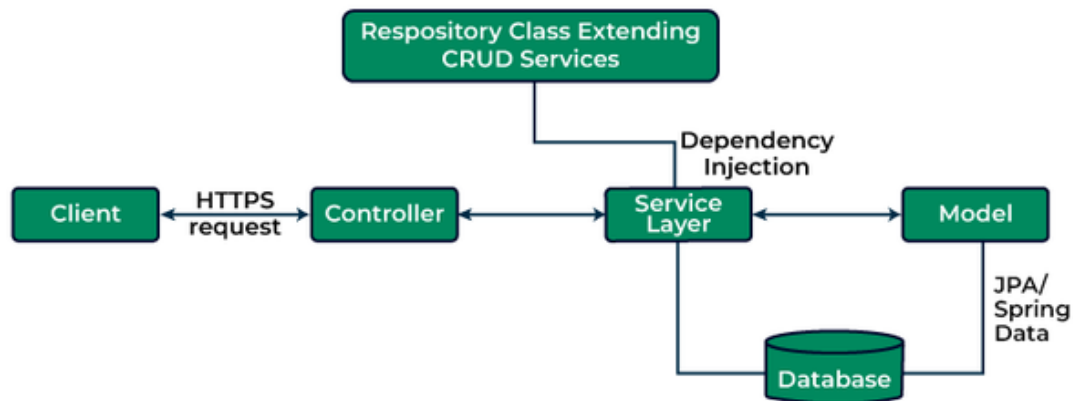
### 5. Model Layer (Entity Layer) -

- Represents database entities and domain models.
- Maps to tables in the database using JPA/Spring Data.

## **6. Database Layer -**

- The actual database that stores application data.
- Spring Boot interacts with it through JPA/Spring Data.

## Request Flow in Spring Boot :-

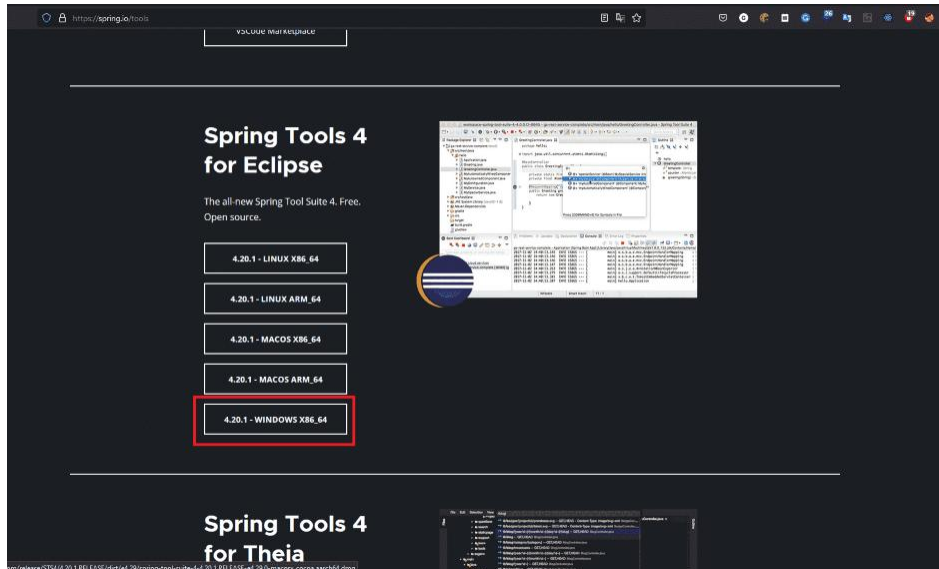


Client ->Controller ->Service ->Repository ->Database ->Response

- A Client makes an HTTPS request (GET/POST/PUT/DELETE).
- The request is handled by the Controller, which is mapped to the corresponding route.
- If business logic is required, the Controller calls the Service Layer.
- The Service Layer processes the logic and interacts with the Repository Layer to retrieve or modify data in the Database.
- The data is mapped using JPA with the corresponding Model/Entity class.
- The response is sent back to the client. If using Spring MVC with JSP, a JSP page may be returned as the response if no errors occur.

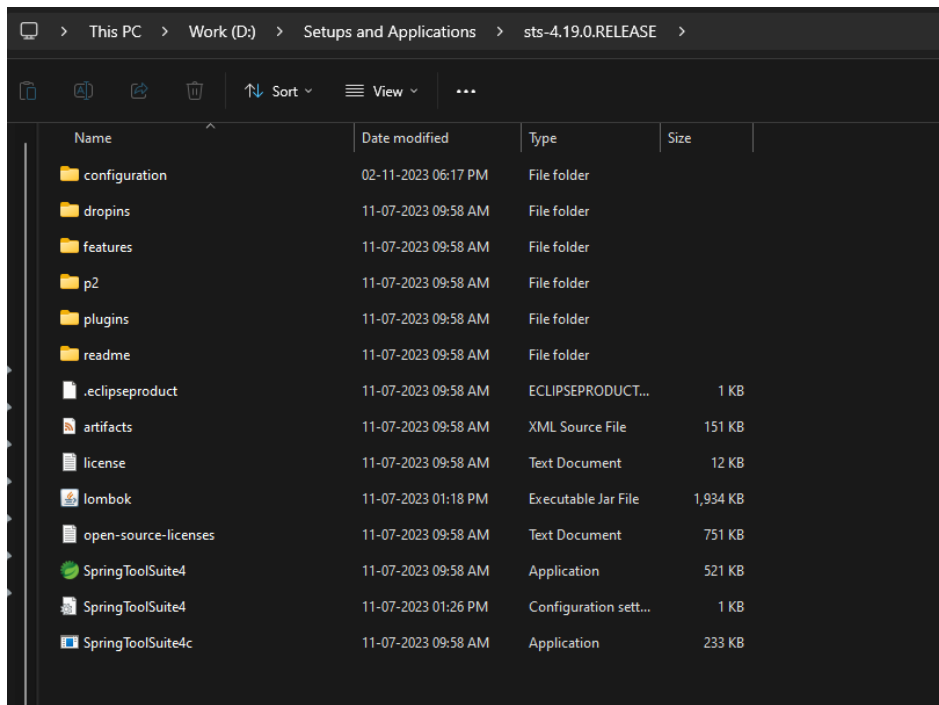
## Installation of Spring Tool Suite on Windows :-

Step 1: Firstly, we need to download the official .zip file of STS from its website. You can download the Spring Tool Suite from the above [link](#). Select the appropriate platform as Windows and download the package.



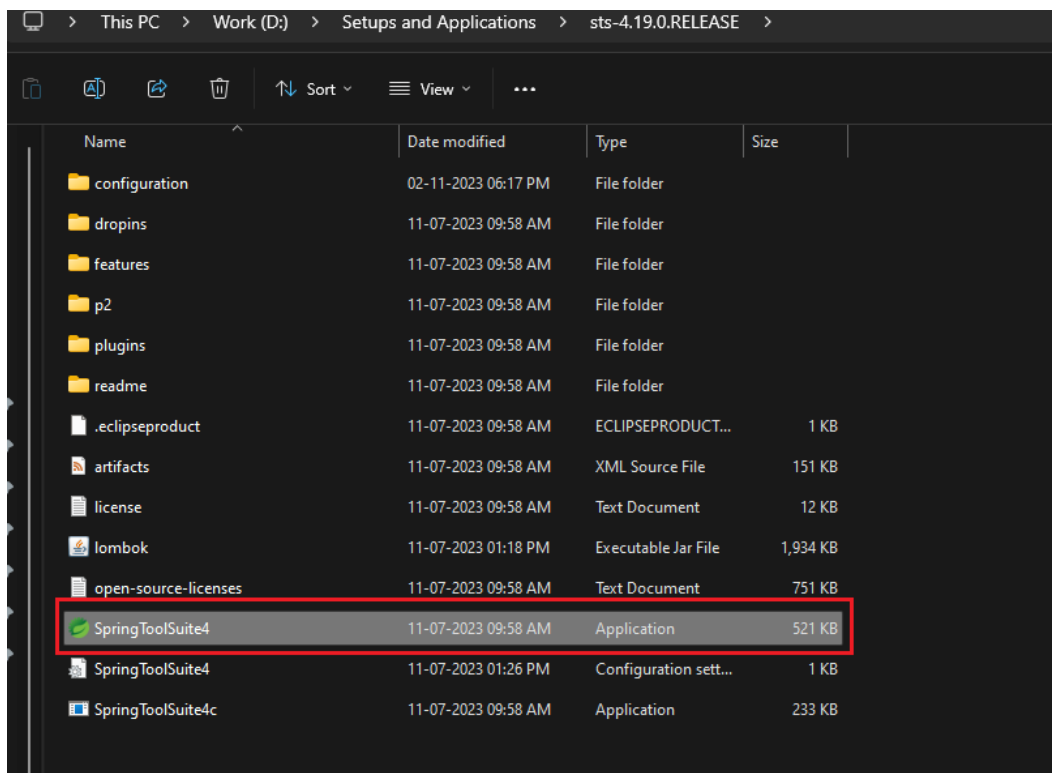
## Downloading STS Zip File

Step 2: Once the download is been done, we need to extract the contents of the ZIP file to the directory of our choice on our Windows Machine.



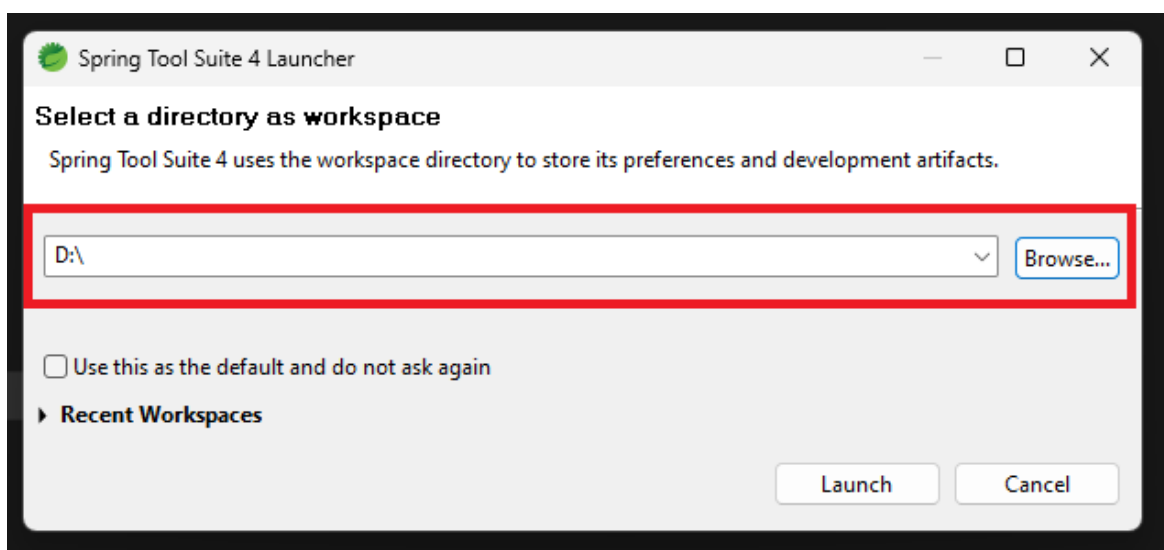
## Extracted Zip File

Step 3: After extracting, we need to navigate to the directory, where we have extracted the STS. In this directory, we will have the file "SprinttoolSuite4". We need to click on that file to open the launcher.



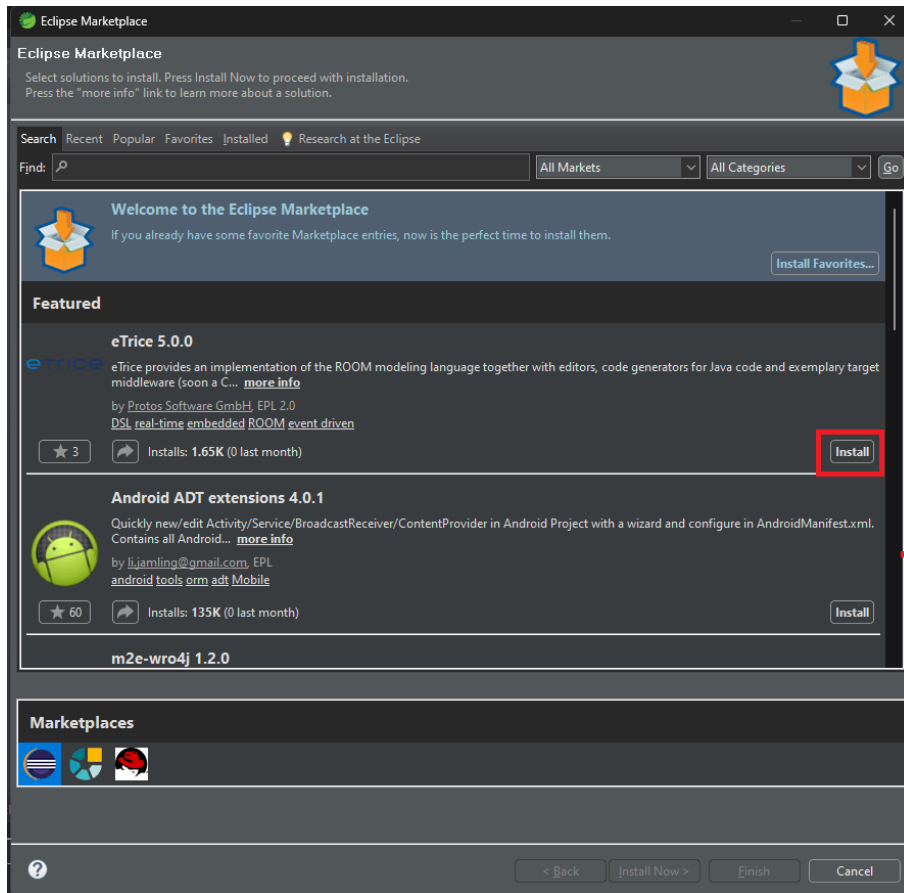
### Clicking on Application

Step 4: Once the launcher is been displayed, we will be prompted to select a workspace location where your projects and configurations will be stored. We can choose the default location or specify your own.



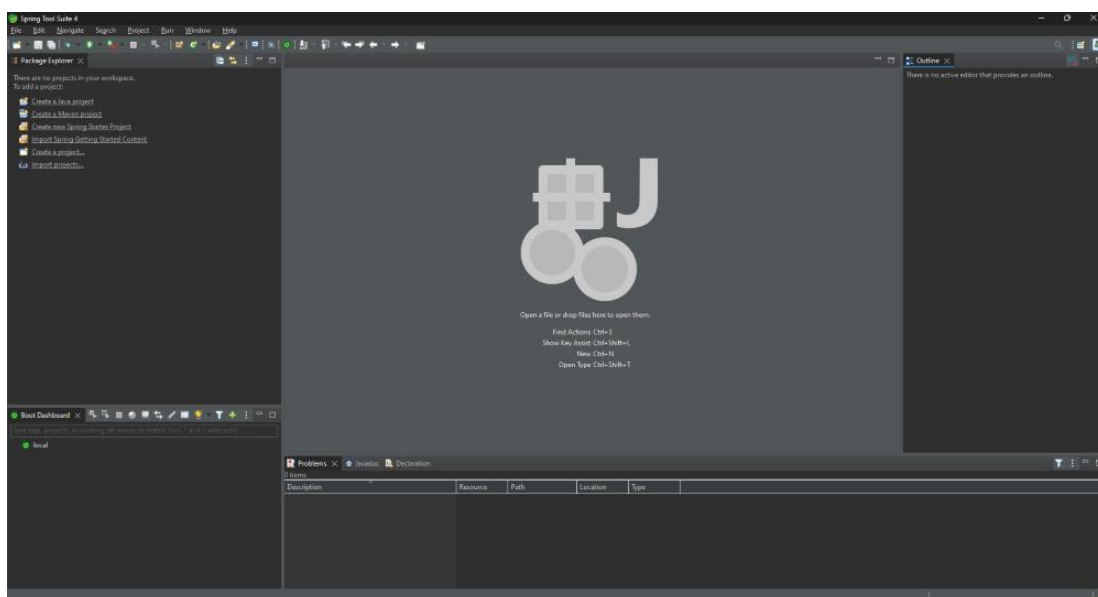
### Setting Up Workspace Path

Step 5: Then we can choose to install additional plugins for specific tasks or frameworks by going to the "Help > Eclipse Marketplace" tab in the STS dashboard.



## Installing Additional Plugins

Step 6: Once the installation is complete, we can start creating and managing Spring-based projects in the Spring Tool Suite IDE.



## Using STS IDE

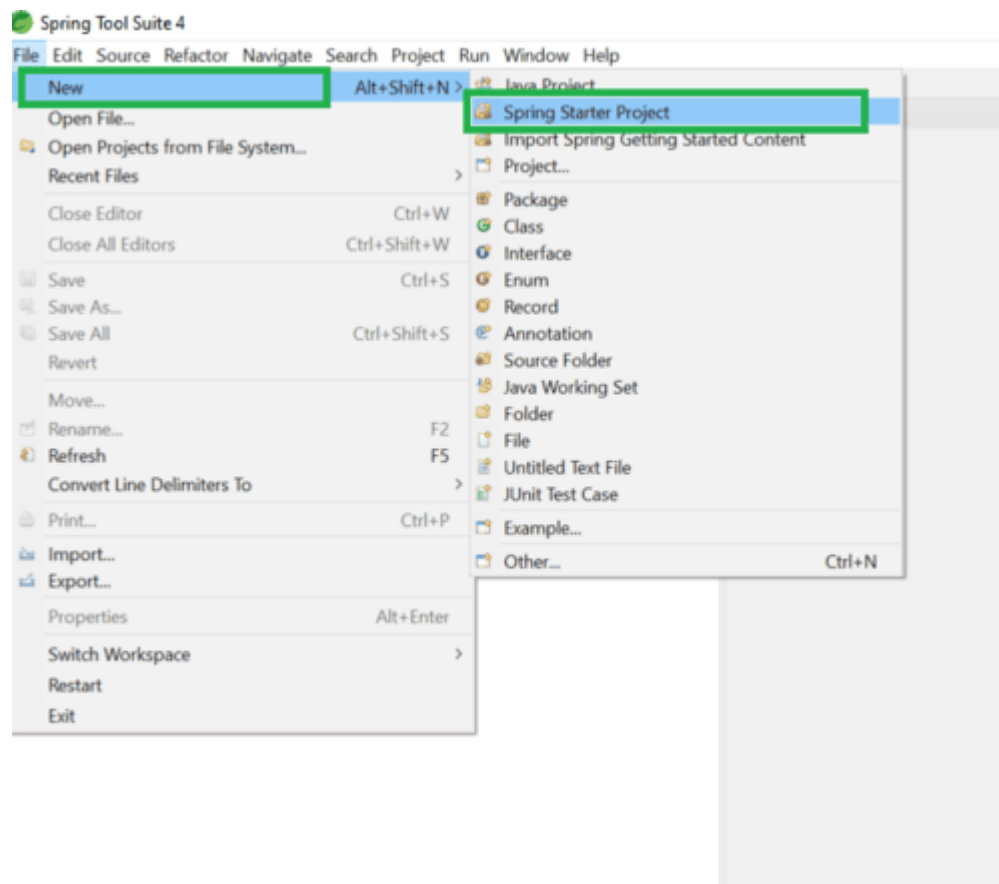
## Create a Spring Boot Project in Spring Tool Suite IDE :-

Spring Tool Suite (STS) is a specialized Eclipse-based IDE designed for Spring applications. It comes with built-in support for Spring Boot, making the setup process faster and easier.

### Step-by-Step Implementation

Step 1: Create a New Spring Starter Project.

- Go to File -> New -> Spring Starter Project
- Enter project details (Name, Group, Artifact, Java version, Packaging, etc.)
- Click Next



STS

New Spring Starter Project



Service URL

https://start.spring.io

Name

demo

☒ Use default location

Location

C:\Users\GFG0574\Documents\workspace-spring-tool-suite-4-4.28.1

Browse

Type:

Maven

Packaging:

Jar

Java Version:

17

Language:

Java

Group

com.example

Artifact

MyProject

Version

0.0.1-SNAPSHOT

Description

Demo project for Spring Boot

Package

com.example.demo

Working sets

☐ Add project to working sets

New...

Working sets:

Select...

?

< Back

Next >

Finish

Cancel

## Spring\_INITIALIZER

New Spring Starter Project Dependencies



Spring Boot Version: 2.5.6

Available:

Type to search dependencies

SQL

☐ JDBC API

☒ Spring Data JPA

☐ Spring Data JDBC

☐ Spring Data R2DBC

☐ MyBatis Framework

☐ Liquibase Migration

☐ Flyway Migration

☐ JOOQ Access Layer

☐ IBM DB2 Driver

☐ Apache Derby Database

☒ H2 Database

☐ HyperSQL Database

☐ MariaDB Driver

☐ MS SQL Server Driver

☐ MySQL Driver

☐ Oracle Driver

☐ PostgreSQL Driver

Security

Spring Cloud

Spring Cloud Circuit Breaker

Selected:

☒ Spring Data JPA

☒ H2 Database

☒ Spring Web

Selected Dependencies

Selected Dependencies

Available Dependency

Make Default

Clear Selection

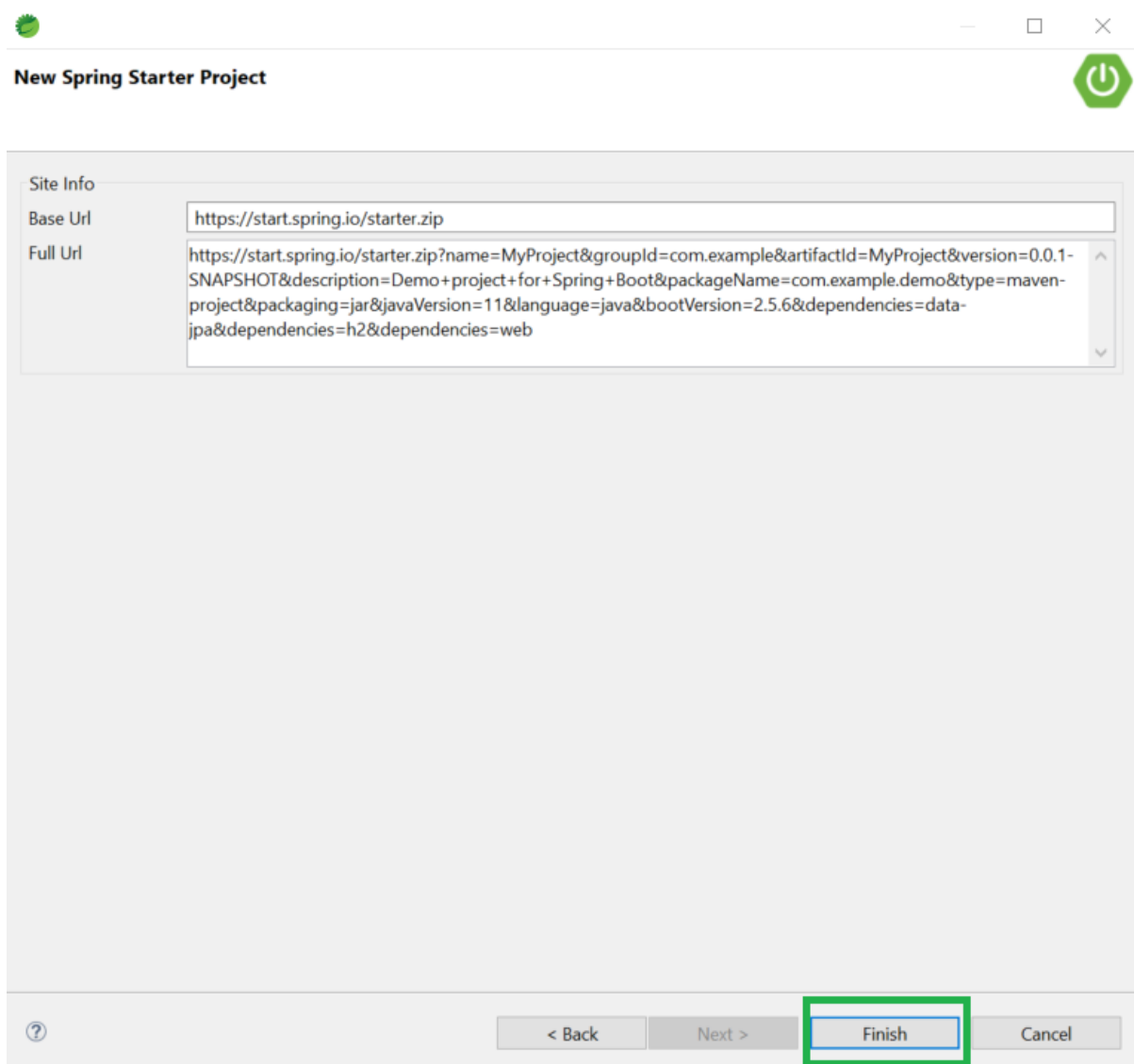
?

< Back

Next >

Finish

Cancel



The image shows a 'New Spring Starter Project' dialog box in a software application. It has a title bar with a green icon and a power button icon. The main area is divided into 'Site Info' and 'Full Url' sections. The 'Base Url' field contains 'https://start.spring.io/starter.zip'. The 'Full Url' field contains a long URL with various parameters: 'https://start.spring.io/starter.zip?name=MyProject&groupId=com.example&artifactId=MyProject&version=0.0.1-SNAPSHOT&description=Demo+project+for+Spring+Boot&packageName=com.example.demo&type=maven-project&packaging=jar&javaVersion=11&language=java&bootVersion=2.5.6&dependencies=data-jpa&dependencies=h2&dependencies=web'. At the bottom, there are four buttons: '< Back', 'Next >', 'Finish' (highlighted with a green box), and 'Cancel'.

**New Spring Starter Project**

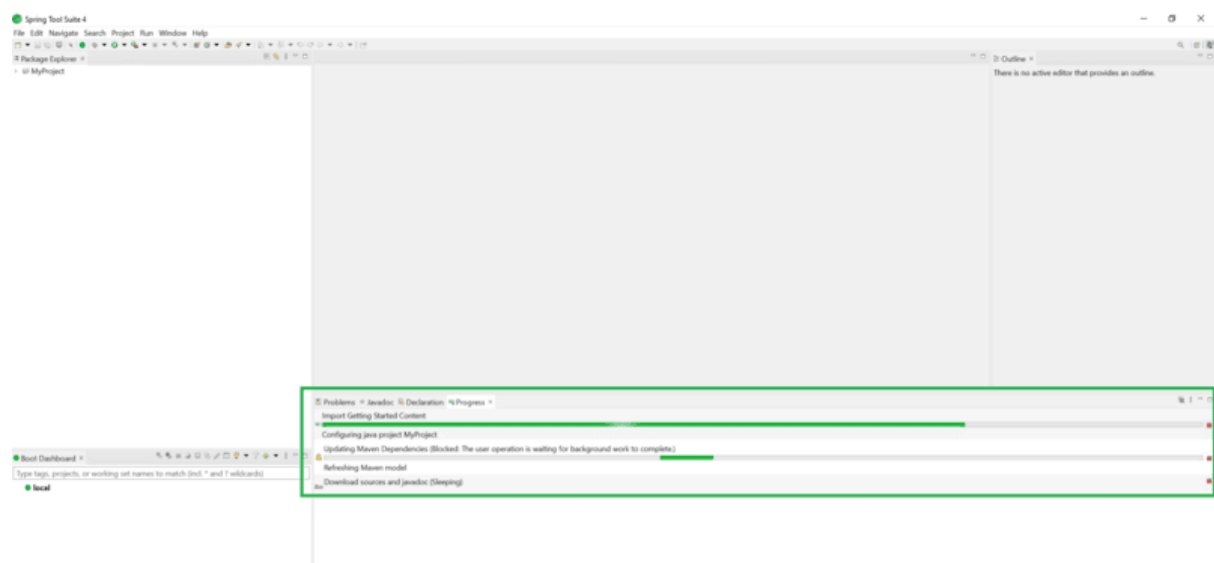
Site Info

Base Url:

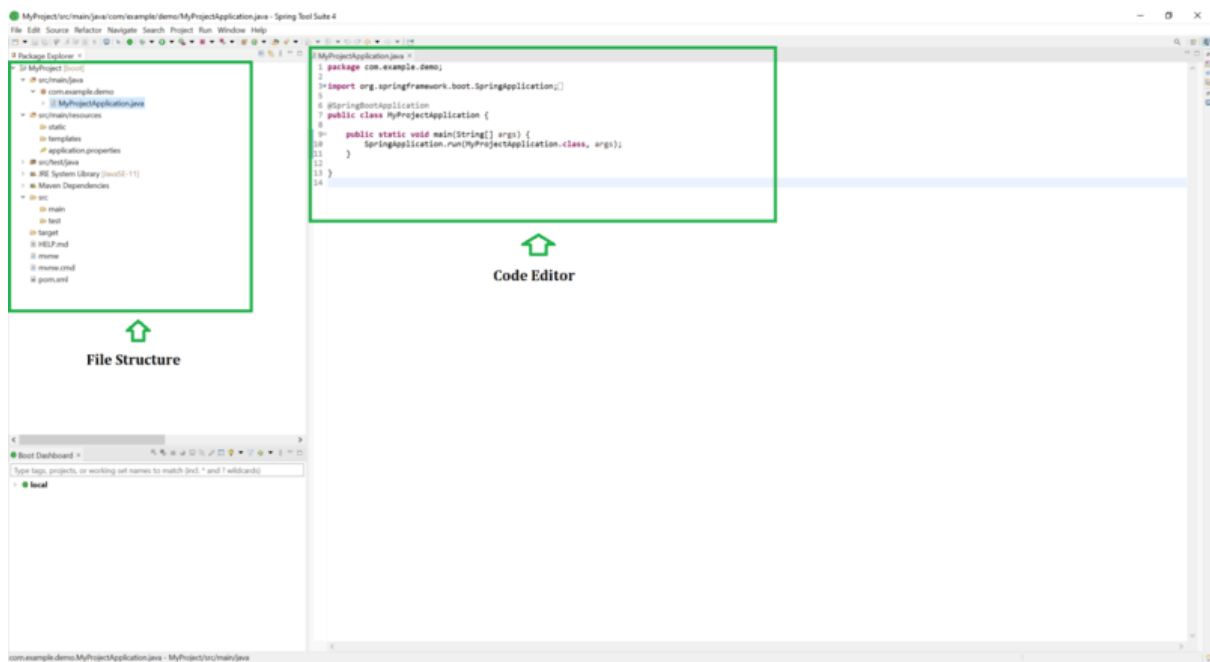
Full Url:

< Back   Next >   **Finish**   Cancel

Here now, please wait for some time to download all the required files such as dependencies that you have selected in Step4 above.



Note: Below is the Welcome screen after you have successfully Created and Setup Spring Boot Project in Spring Tool Suite.



## Implementation of First Project of Spring Boot :-

### Add Dependencies to pom.xml –

```
https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>4.0.0</version>
    <relativePath/> <!-- lookup parent from repository -->
  </parent>
  <groupId>com.example</groupId>
  <artifactId>Spring_Boot_Sample_Project1</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>Spring_Boot_Sample_Project1</name>
  <description>Demo project for Spring Boot</description>
  <url/>
  <licenses>
    <license/>
  </licenses>
  <developers>
    <developer/>
  </developers>
  <scm>
    <connection/>
    <developerConnection/>
    <tag/>
    <url/>
  </scm>
  <properties>
    <java.version>17</java.version>
  </properties>
  <dependencies> Add Spring Boot Starters...
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-webmvc</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-devtools</artifactId>
      <scope>runtime</scope>
      <optional>true</optional>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-webmvc-test</artifactId>
      <scope>test</scope>
    </dependency>

    <!-- https://mvnrepository.com/artifact/org.apache.tomcat.embed/tomcat-embed-jasper -->
    <dependency>
      <groupId>org.apache.tomcat.embed</groupId>
      <artifactId>tomcat-embed-jasper</artifactId>
      <scope>provided</scope>
    </dependency>
```

```

<!-- jsp pages-->
<!-- https://mvnrepository.com/artifact/jakarta.servlet.jsp.jstl/jakarta.servlet.jsp.jstl-api -->
<dependency>
    <groupId>jakarta.servlet.jsp.jstl</groupId>
    <artifactId>jakarta.servlet.jsp.jstl-api</artifactId>
</dependency>

<dependency>
    <groupId>org.glassfish.web</groupId>
    <artifactId>jakarta.servlet.jsp.jstl</artifactId>
</dependency>

</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>

</project>

```

#### SpringBootSampleProject1Application.java –

```

package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringBootSampleProject1Application {

    public static void main(String[] args) {
        SpringApplication.run(SpringBootSampleProject1Application.class, args);
    }

}

```

#### MyController.java –

```

package com.example.demo;

import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;

@Controller
public class MyController {

    @GetMapping("/")
    public String home()
    {
        return "home";    //home.jsp
    }

}

```

**application.properties –**

```
spring.application.name=Spring_Boot_Sample_Project1
```

```
spring.mvc.view.prefix=/WEB-INF/pages/
```

```
spring.mvc.view.suffix=.jsp
```

**home.jsp –**

```
<%@ page language="java" contentType="text/html; charset=UTF-8"  
    pageEncoding="UTF-8"%>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Insert title here</title>
```

```
</head>
```

```
<body>
```

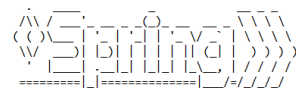
```
    <h2>Welcome To Spring Boot</h2>
```

```
    <!-- webapp/WEB-INF/pages/home.jsp -->
```

```
</body>
```

```
</html>
```

### Console Based –



```

:: Spring Boot ::               (v4.0.0)

2025-11-26T18:29:12.759+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] e.d.SpringBootSampleProjectApplication : Starting SpringBootSampleProjectApplication using Java 21.0.8 with
PID 12004 (C:\Users\sagar\workspace\workspace\Spring_Boot_Sample_Project\target\classes started by sagar in C:\Users\sagar\workspace\workspace\Spring_Boot_Sample_Project)
2025-11-26T18:29:12.766+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] e.d.SpringBootSampleProjectApplication : No active profile set, falling back to 1 default profile: "default"
2025-11-26T18:29:12.869+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-
properties' to 'false' to disable
2025-11-26T18:29:12.870+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the
'spring.level.web' property to 'DEBUG'
2025-11-26T18:29:12.894+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2025-11-26T18:29:12.944+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2025-11-26T18:29:12.944+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.14]
2025-11-26T18:29:13.179+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] org.apache.jasper.servlet.TldScanner : At least one JAR was scanned for TLDs yet contained no TLDs. Enable
debug logging for this logger to see the list of JARs that were scanned but no TLDs were found in them. Skipping unnecessary JARs during
2025-11-26T18:29:15.193+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : WebApplicationContext: initialization completed in 2322 ms
2025-11-26T18:29:15.975+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2025-11-26T18:29:15.986+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [ restartedMain] e.d.SpringBootSampleProjectApplication : Started SpringBootSampleProjectApplication in 4.113 seconds
(process running for 5.731)
2025-11-26T18:29:58.452+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2025-11-26T18:29:58.453+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [nio-8080-exec-1] s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2025-11-26T18:29:58.455+05:30 INFO 12004 --- [Spring_Boot_Sample_Project] [nio-8080-exec-1] s.web.servlet.DispatcherServlet : Completed initialization in 2 ms

```

### Web Based –



# Wecome To Spring Boot